Aarthi Venkat, Ph.D.

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CONTACT INFORMATION

linkedin.com/in/aarthi-venkat/ (408) 799-9189
aarthivenkat.github.io avenkat@broadinstitute.org

EDUCATION

Yale University
Ph.D. in Computational Biology & Bioinformatics — May 2024
M.S. in Computational Biology & Bioinformatics — Dec 2021

The University of California, San Diego
B.S. in Bioengineering: Bioinformatics — Jun 2019

RESEARCH EXPERIENCE

Postdoctoral Associate — Sept 2024-Present
Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard
& Harvard Medical School, Drs. Nir Hacohen & Marinka Zitnik

- Build independently-funded program designing graph-based approaches for transfer learning in functional genomics across modalities and biological scales
- Collaborate with researchers at the Broad Institute and Roche Pharmaceuticals toward driving actionable, patient-centered insights

Computational Biology & Bioinformatics Ph.D. — Aug 2019-Jul 2024
Yale University, Dr. Smita Krishnaswamy

- Developed 7 geometric representation learning methods for single-cell and spatial sequencing data to reveal underlying cellular and molecular behavior
- Co-led 8 collaborations across diverse biological disciplines (e.g. immunology, cancer, genetics) to build integrative frameworks for scientific discovery

Applied Science Research Intern — Sept 2021-Dec 2021
Google Brain, Drs. Lucy Colwell & Farhad Hormozdiari

- Designed machine learning-guided directed evolution approach for CRISPR guide RNA design, outperforming baselines on efficacy and expression prediction
- Worked closely with researchers at Google Genomics and UCSF to validate findings

Bioinformatics Research Assistant — Oct 2016-Sept 2019
La Jolla Institute for Immunology, Dr. Ferhat Ay

- Elucidated the 3D genome from Hi-C sequencing of 7 apicomplexan parasite species
- Corrected *Toxoplasma gondii* misassembly with Hi-C and long-read sequencing

Computational Biology Research Assistant — Sept 2018-Aug 2019
Institute for Genomic Medicine, Dr. Theresa Gaasterland

- Performed bioinformatic analysis of genetic variation in primary congenital glaucoma

Genome Informatics Intern — Jun 2018-Sept 2018
Regeneron Genetics Center, Regeneron Pharmaceuticals

- Integrated loss-of-function variants from 500,000+ exomes for target priorization
- Selected as **1 of 7 out of 250+ interns** to give research talk at company-wide event

Data Analytics Intern — Jun 2017-Sept 2017
Research & Development, Auris Health

- Built cloud-based pipeline to facilitate high-performance analysis of endoscopic robot

SELECTED PUBLICATIONS

*§ Equal contribution.
Links to full articles at: `https://aarthivenkat.github.io`

[1] **A Venkat**, Z Jiang*, D Marbach*, N Hacohen§, M Zitnik§. *Cross-modal transfer learning for mapping bulk transcriptomes at cellular level* (2026).
– Presented at ICML 2026 Multi-modal Foundation Models and Large Language Models for Life Sciences Workshop, EWSC Symposium 2026: Biomedical Science and AI

[2] C Garcia*, **A Venkat***... S Krishnaswamy§, MD Muzumdar§. *Beta cell-derived cholecystokinin drives obesity-associated pancreatic adenocarcinoma development.* Nature Communications (2026).
– Presented at WNAR/IMS 2026, ISMB MLCSB 2025, AACR 2022/2024/2025, Yale Single Cell Symposium 2022, Broad Institute Symposium 2024, Dr. Samuel M. Nabrit Conference for Early Career Scholars 2025
– Followed up by McQuaid et al. (AACR 2026, effect of treatment with GLP1-RAs)

[3] **A Venkat***, S Youlten*, BP San Juan*... S Krishnaswamy§, CL Chaffer§. *AAnet resolves a continuum of spatially-localized cell states to unveil tumor complexity.* Cancer Discovery (2025).
– **Featured selection by AACR Publications** at 2026 AACR Meeting
– Presented at WNAR/IMS 2026, University of Idaho Colloquium 2026, CSHL Systems Immunology 2021
– Followed up by Garcia, Venkat, McQuaid et al. (Nature Communications, β cell archetypes), Torices et al. (In progress, atlas-level archetypal analysis)

[4] **A Venkat**, S Leone, S Youlten, E Fagerberg, J Attanasio, NS Joshi, S Krishnaswamy. *Mapping the gene space at single-cell resolution with gene signal pattern analysis.* Nature Computational Science (2024).
– Presented at University of Idaho Colloquium 2026, Learning on Graphs 2025, Yale Genetics Symposium 2023, Gruber Symposium 2023, AAI 2023, ISMB MLCSB 2021
– Followed up by Thomas et al. in trainee-led work (ASCB 2025, patient-specific gene-gene networks)

Journal Articles

[5] **A Venkat***, M Carlino*, B Lawton*... S Krishnaswamy§, D Krause§. *Single-cell analysis reveals transcriptional dynamics in primary parathyroid tissue.* Genome Research (2024).

[6] **A Venkat**, D Bhaskar, S Krishnaswamy. *Multiscale geometric and topological analyses for characterizing and predicting immune responses from single-cell data.* Cell Trends in Immunology (2023).

[7] M Damo, N Hornick, **A Venkat**... NS Joshi. *PD-1 prevents pathogenicity of effector CD8 T cells that infiltrate skin under homeostatic conditions.* Nature (2023).

[8] M Amodio, SE Youlten, **A Venkat**, BP San Juan, CL Chaffer, S Krishnaswamy. *Single-cell multi-modal GAN reveals spatial patterns in single-cell data from triple-negative breast cancer.* Cell Patterns (2022).

[9] KA Connolly, M Kuchroo, **A Venkat**... NS Joshi. *A reservoir of stem-like CD8+ T cells in the tumor draining lymph node preserves the ongoing antitumor immune response.* Science Immunology (2021).

[10] Y Su*, **A Venkat***, Y Yadav, L Puglisi, S Fodeh. *Twitter-based analysis reveals differential COVID-19 concerns across areas with socioeconomic disparities.* Computers in Biology and Medicine (2021).

[11] J Xia, **A Venkat**, ML Reese, KG Le Roch, F Ay, JP Boyle. *Third generation sequencing revises the molecular karyotype for Toxoplasma gondii and identifies emerging copy number variants in sexual recombinants.* Genome Research (2021).

[12] EM Bunnik, **A Venkat***, J Shao*, KE McGovern... F Ay§, KG Le Roch§. *Comparative 3D Organization in Apicomplexan Parasites.* Proceedings of the National Academy of Sciences (2019).

Conference Proceedings

[13] **A Venkat***, J Chew*, F Cardoso Rodriguez, CJ Tape, M Perlmutter§, S Krishnaswamy§. *Directed scattering for knowledge graph-based cellular signaling analysis.* ICASSP (2024).

[14] D Bhaskar*, DS Magruder*, M Morales, E De Brouwer, **A Venkat**, F Wenkel, J Noonan, G Wolf, N Ivanova, S Krishnaswamy. *Inferring dynamic regulatory interaction graphs from time series data with perturbations.* LoG Conference (2023).

[15] S Leone, A Tong, G Huguet, **A Venkat**, G Wolf, S Krishnaswamy. *Graph Fourier MMD for Signals on Graphs.* SampTA (2023).

Preprints & In Review

[16] X Sun*, S Gupta*, A Tong*, M Kuchroo*, D Bhaskar*... **A Venkat**... CL Chaffer§, S Krishnaswamy§. *Revealing dynamic temporal trajectories and underlying regulatory networks with Cflows.* In Revision at Nature Genetics.

[17] V Giunchiglia*, O Queen*... **A Venkat**... M Zitnik. *CASCADE: context-aware single-cell modelling links cellular programmes to patient-level disease phenotypes.*

Invited & Award Talks

- *Geometric machine learning enables multiscale perturbational reasoning*
  Stellar Abstract Award, Harvard T.H. Chan Annual PQG Conference (2026)
- *Dissecting single-cell biology across contexts with geometric machine learning*
  University of Massachusetts Lowell (2026)
- *Uncovering cellular mechanisms of tumorigenesis with archetypal analysis*
  Western North American Region Institute of Mathematical Statistics Meeting (2026)
- *Decoding cellular biology with geometric machine learning*
  University of Idaho Math & Statistics Colloquium (2026)
- *Geometric representation learning for single-cell biology across contexts*
  Singh Lab, Brown University (2025)

Oral Presentations

- *Semantic Analysis Reveals Increasing Attribution of Responsibility to AI in Scientific Literature*, Women in Bioinformatics Workshop [Presented by Nic Fisk] (2026)
- *Relational ethics for scientific knowledge production and the limits of LLMs*
  Machine Logos: Persons, Language, and AI (2026)
- *Epistemic Responsibility and Interdisciplinarity in the Age of AI for Science*
  LLMs and Digital Autonomy: From Misinformation to Context Collapse (2025)
- *Dissecting cellular and molecular mechanisms of pancreatic cancer with deep learning*
  Intelligent Systems for Molecular Biology (ISMB) Machine Learning in Computational and Systems Biology Track (2025)
- *Dissecting cellular and molecular mechanisms of pancreatic cancer*
  Dr. Samuel M. Nabrit Conference, Brown University (2025)
- *Learning directed and hyperbolic embeddings*
  Graph Signal Processing Workshop (2023)
- *PHATE reveals cell state transformation in Tercen biomedical data analysis platform*
  CYTO Conference (2023)
- *Elucidating mechanisms of endocrine-exocrine signaling in pancreatic cancer*
  Yale Single Cell Symposium (2022)
- *Leveraging the Power of Human Genetics through Knockout Discovery*
  Regeneron Pharmaceuticals Annual Meeting (2018)

Poster Presentations

- *Cross-modal transfer learning for mapping bulk transcriptomes at cellular level*
  ICML 3rd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences (2026)
- *Cross-modal transfer learning for mapping bulk cancer phenotypes at cellular resolution*, Eric and Wendy Schmidt Symposium: Biomedical Science and AI (2026)
- *Analysis of Therapeutic Response in Hepatocellular Carcinoma with Graph-based Machine Learning Methods on Spatial Transcriptomics Data*
  American Society for Cell Biology (ASCB) [Presented by Hannah Thomas] (2025)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
  Learning on Graphs (LOG) (2025)
- *Learning cellular and molecular mechanisms of pancreatic cancer*
  Connecting the Dots Broad Institute Symposium (2024)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
  Yale Department of Genetics Symposium (2023)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
  Gruber Science Fellowship Symposium (2023)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
  The American Association of Immunologists Conference (2023)

- *Manifold-based gene density estimates reveal immune signaling in meningioma*
  Intelligent Systems for Molecular Biology (ISMB) Machine Learning in Computational and Systems Biology Track (2021)
- *Archetypal analysis of antigen-specific T cell responses across conditions*
  Cold Spring Harbor Laboratory Systems Immunology Conference (2021)

Teaching Experience

Teaching Fellow, Deep Learning Theory and Applications S 2021, S 2024
Yale University, Computer Science
- Designed scaffolded and interactive materials for building deep learning foundations
- Mentored groups to extend and apply deep learning to real-world application areas

Student Advisory Board, Poorvu Teaching Center 2023
- Advised on institutional curricula/policies integrating AI literacy with DEI principles

Teaching Assistant, Computational Genomics Nov 2022, Dec 2023
Cold Spring Harbor Laboratory Workshop
- Designed presentations and guided coding notebooks on machine learning for single-cell analysis for 50 PhD-level researchers across all levels
- Trained researchers to perform independent analysis of their own datasets

Teaching Assistant, Machine Learning for Single-cell Analysis May 2020, Jan 2021
Yale University, Department of Genetics & Yale SEAS
- Co-taught **100+ researchers** in single-cell analysis (`krishnaswamylab.org/workshop`)

Teaching Assistant, Introduction to Biomedical Data Science and Health Informatics Jun 2020
Yale Center for Medical Informatics
- Trained researchers across all career stages in practical biomedical data analysis skills

Undergraduate Tutor / Instructional Assistant, Genetics F 2017, F 2018, S 2019
UC San Diego Biological Sciences
- Developed material for student-centered learning in recitations and office hours
- Received **Excellence in Teaching Award** for top evaluation (100% positive reviews)

Fellowships & Grants

Eric and Wendy Schmidt Center Fellowship Sep 2025
Broad Institute of MIT and Harvard
- Selectively awarded to train next generation of leaders in machine learning and biology

Graduate Student Assembly Conference Travel Fellowship Jun 2023, Mar 2024
Yale University

2021-2023
Yale Cancer Biology Training Grant
Yale School of Medicine
- Awarded to next generation of cancer scientific leaders to provide unique training in cancer biology and practical clinical issues in oncology

Yale Gruber Science Fellowship Aug 2019
Yale University
- Awarded the highest honor for incoming science PhDs by Graduate School of Arts and Sciences in recognition of outstanding accomplishments and promise

Honors & Awards

Stellar Abstract Award 2026
Harvard T.H. Chan School of Public Health
- Awarded to stellar abstracts for presentation at Harvard T.H. Chan PQG conference

Dr. Samuel M. Nabrit Conference for Early Career Scholars 2025
Molecular Biology, Cell Biology, and Biochemistry, Brown University
- Showcasing outstanding molecular life scientists who have worked to advance the role of historically underrepresented groups

Public Communication Certificate 2023
Poorvu Center for Teaching & Learning, Yale University
- Completion of program training to communicate complex research to broad audience

Finalist for Yale Research Excellence Award 2022
Yale School of Medicine, Office of Health Equity Research
- Received for "Twitter-based analysis reveals differential COVID-19 concerns across areas with socioeconomic disparities"

Outstanding Academic Achievement in Bioengineering 2019
The University of California, San Diego
- Highest GPA (**1st rank**) in graduating class in Bioengineering: Bioinformatics

Excellence in Teaching Award 2019
The University of California, San Diego
- Highest performance evaluation for teaching assistance in Genetics

Tau Beta Pi Engineering Honors 2018, 2019
The University of California, San Diego
- Awarded to engineering students displaying high academic achievement and integrity

Muir College Caledonian Honors 2018, 2019
The University of California, San Diego
- Awarded to senior students displaying outstanding academic achievement (>3.8 GPA)

Provost Honors 2015-2019
The University of California, San Diego
- Received 12 times (all academic years) for high academic achievement (> 3.5 GPA)

Academic Service

Invited Reviewer and PC Member 2023-2026
- Reviewed and co-reviewed papers for Nature, BMC Bioinformatics, Computational & Structural Biotechnology Journal, Yale Journal of Biology & Medicine, ICML GFM 2026, ICML GenBio 2026, ICLR Gen^2 2026, ISMB 2026, ISMB 2025, RECOMB 2024

Research Mentor 2023-2026
- Katherine Wang, MIT Computer Science
- Dominique Dang, MIT Computer Science & Molecular Biology
- Hannah Thomas, Nashua High School South → Research Science Institute @ MIT
- Sofia Lara, MIT Biological Engineering → Rhodes Scholar PhD @ Oxford University
- Sam Leone, Yale Applied Mathematics → MIT Lincoln Laboratory

Cambridge School Volunteers (KeyPals) 2026
- Fostered scientific career awareness through volunteering with local elementary schools

Broad Summer Research Program Selection Committee 2025
- Selected underrepresented undergraduate students with strong interest in genomics for contributing summer research at the Broad Institute

Foundation Models in Genomics Panel Moderator @ Broad 2024
- Led discussion on opportunities, successes, and pitfalls of biological foundation models

Networking Chair, Yale Gruber Science Fellowship 2022, 2023
- Hosted networking talks, panels, and discussion to foster Gruber scientific community

Student Representative, Graduate Student Assembly 2023
- Advocated for CB&B graduate students to improve Yale healthcare literacy and policy

Reviewing Editor, Yale Journal of Biology & Medicine 2023
- Made editorial decisions and navigated rounds of review for *Big Data* issue

Social Services & Insurance Counseling, HAVEN Free Clinic 2022, 2023
- Provided healthcare guidance and resources to uninsured New Haven residents

Cancer Biology Training Program, Yale School of Medicine 2021-2023
- Shadowed Dr. Pamela Kunz and discussed clinical relevance of *in silico* cancer research